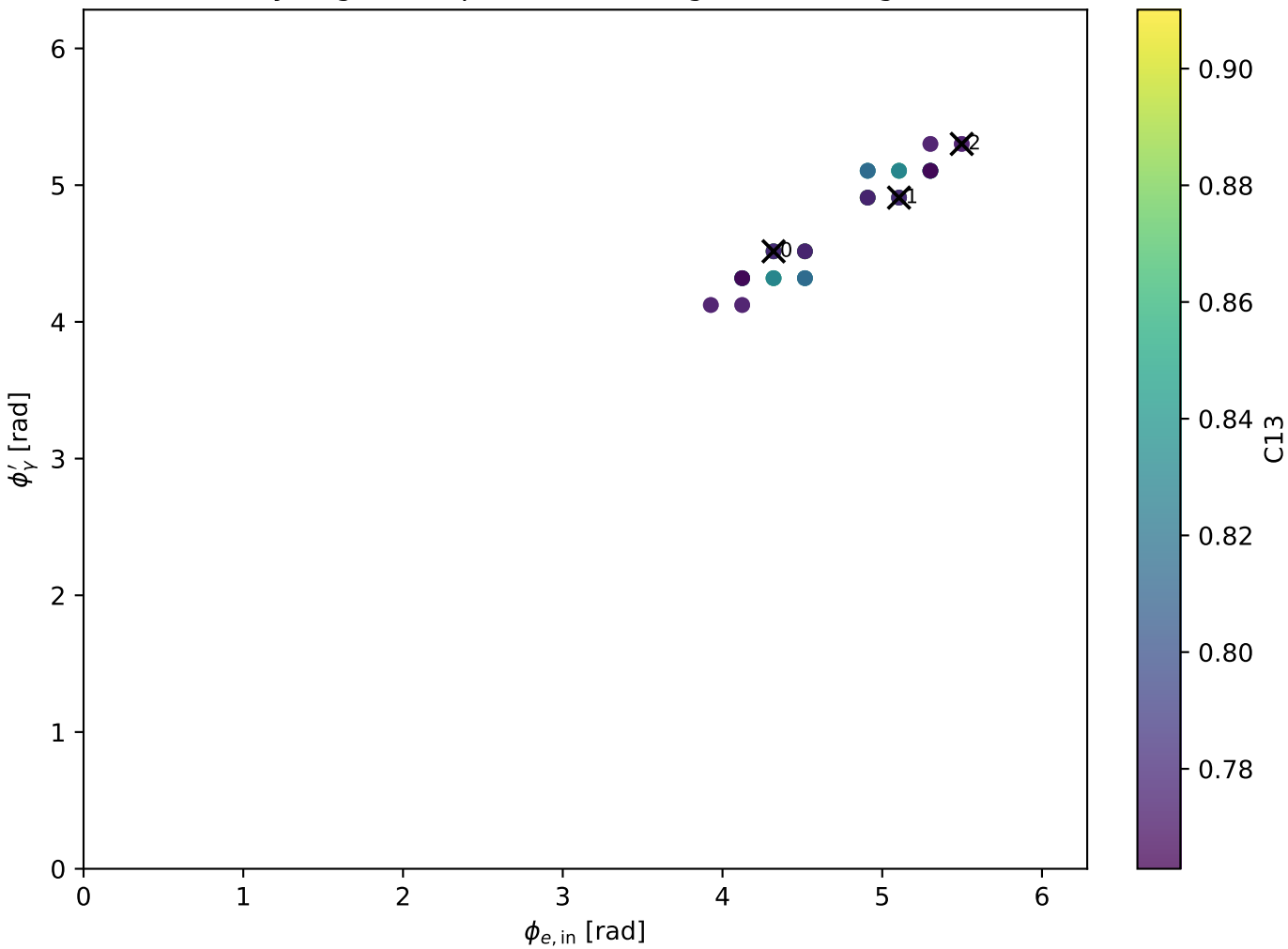
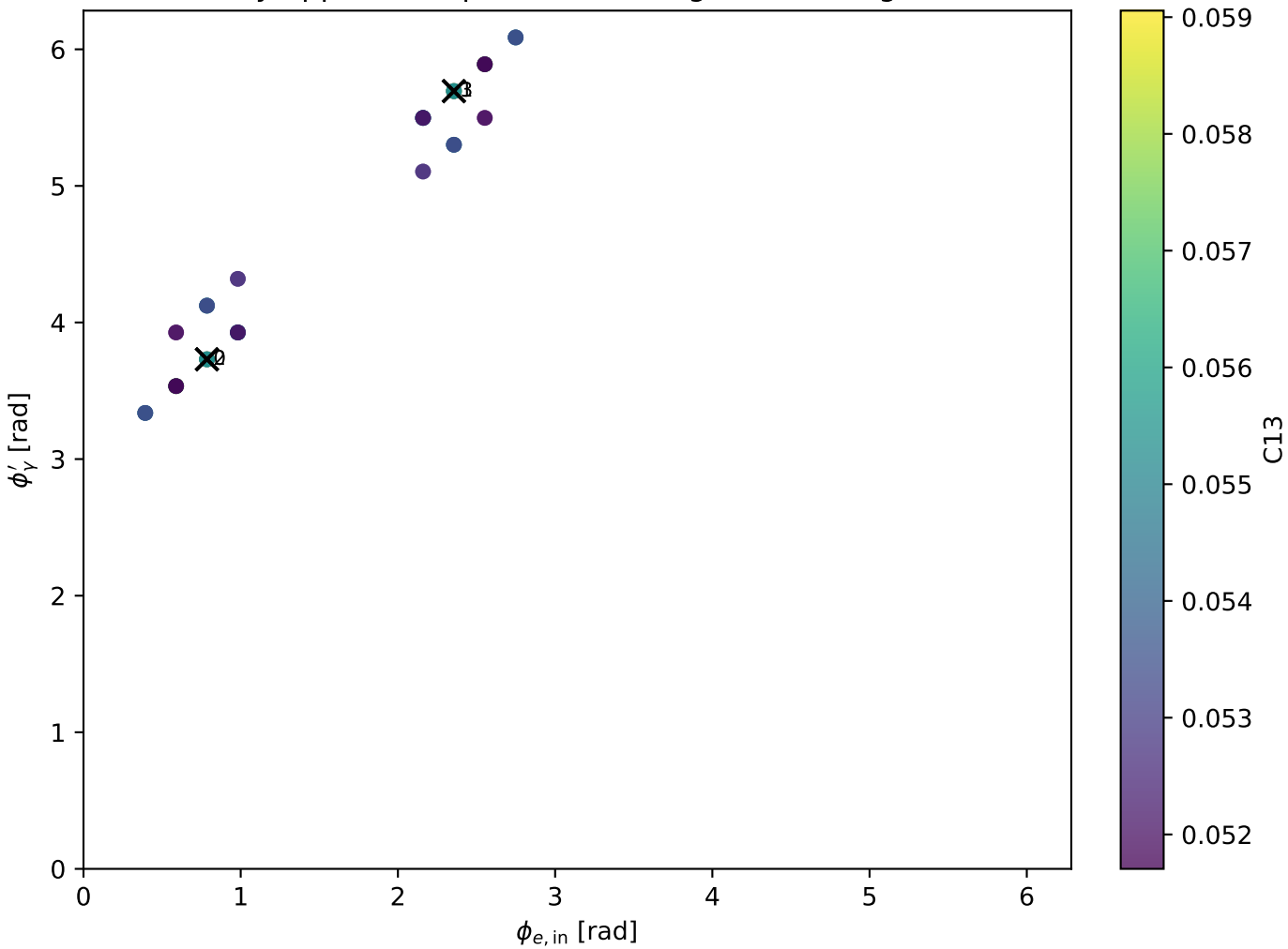


Ty aligned: representative high-C13 configs

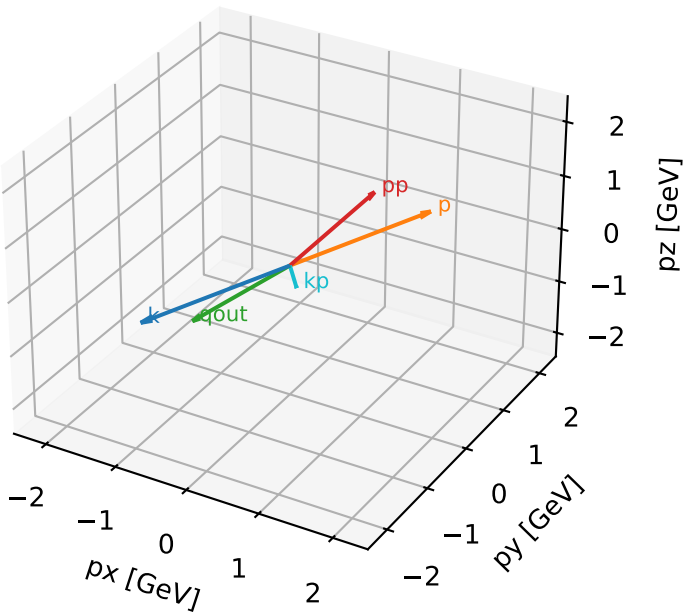


Ty opposite: representative high-C13 configs



Ty characteristic high-C13 configuration

3D momenta

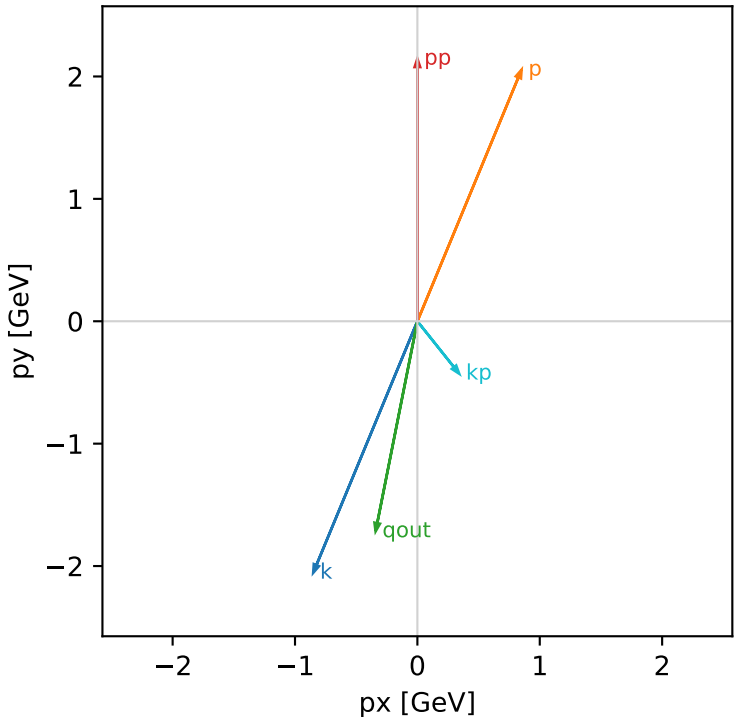


Ty_aligned_cluster_0 C13=0.910096
region: aligned $|\phi_{e,in} - \phi_{\gamma}|=0.19635$
kinematic point: mid_s_low_theta_in_high_Egamma
incoming state: $(A[h_{In}=+1,s_{In}=1] + i A[h_{In}=-1,s_{In}=1]) / \sqrt{2}$

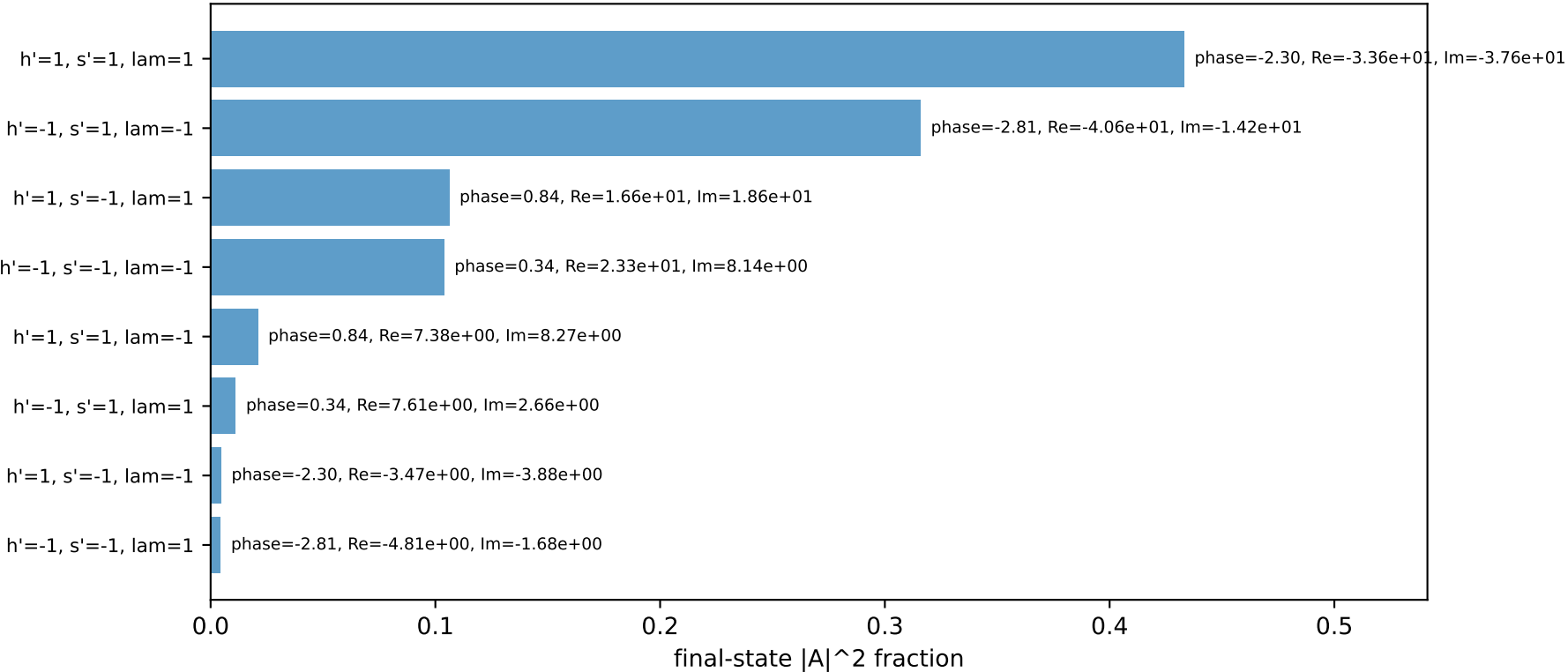
$s=21.5158$, $\sqrt{s}=4.63852$, $p_{In}=2.22442$, $p_{Out}=2.14466$
 $\theta_{in}=1.57079$, $\phi_{e,in}=4.31969$, $\phi_{p,in}=1.1781$
 $q_{Out}=1.75$, $\phi_{\gamma}=4.51604$
 $Q^2=1.25759$, $x_B=0.0748011$, $t=-0.727273$, $W_2=16.4347$, $y=0.814716$
 $\theta(e',\gamma)=49.8104$ deg

four-momenta [E, p_x , p_y , p_z] GeV:
k E= 2.2244 $p_x=-0.8512$ $p_y=-2.0551$ $p_z=-0.0000$
p E= 2.4141 $p_x=0.8512$ $p_y=2.0551$ $p_z=0.0000$
kp E= 0.5477 $p_x=0.3414$ $p_y=-0.4283$ $p_z=0.0000$
pp E= 2.3408 $p_x=0.0000$ $p_y=2.1447$ $p_z=0.0000$
qout E= 1.7500 $p_x=-0.3414$ $p_y=-1.7164$ $p_z=0.0000$

Transverse plane

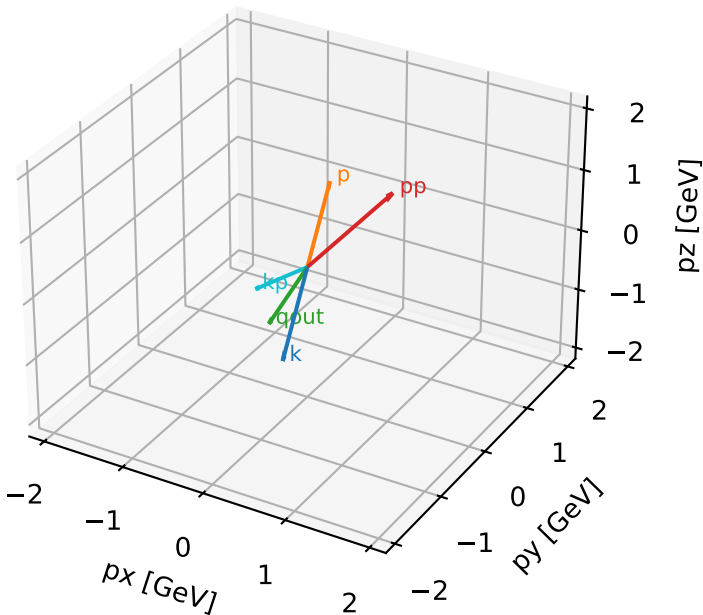


Final-state amplitude decomposition



Ty characteristic high-C13 configuration

3D momenta

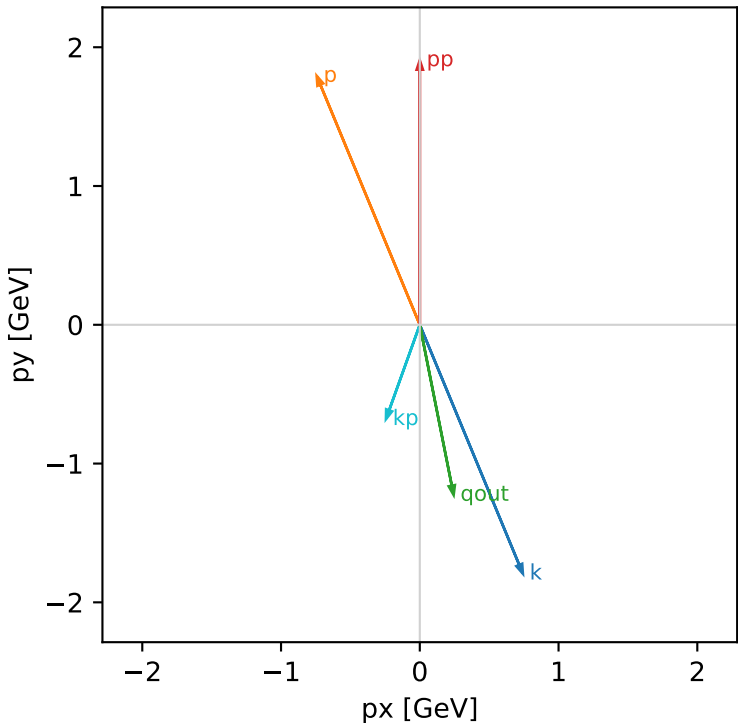


Ty_aligned_cluster_1 C13=0.777558
region: aligned $|\phi_{e\text{ in}} - \phi_{\gamma}|=0.19635$
kinematic point: low_s_low_theta_in_mid_Egamma
incoming state: $(A[h_{In}=+1,s_{In}=1] + i A[h_{In}=-1,s_{In}=1]) / \sqrt{2}$

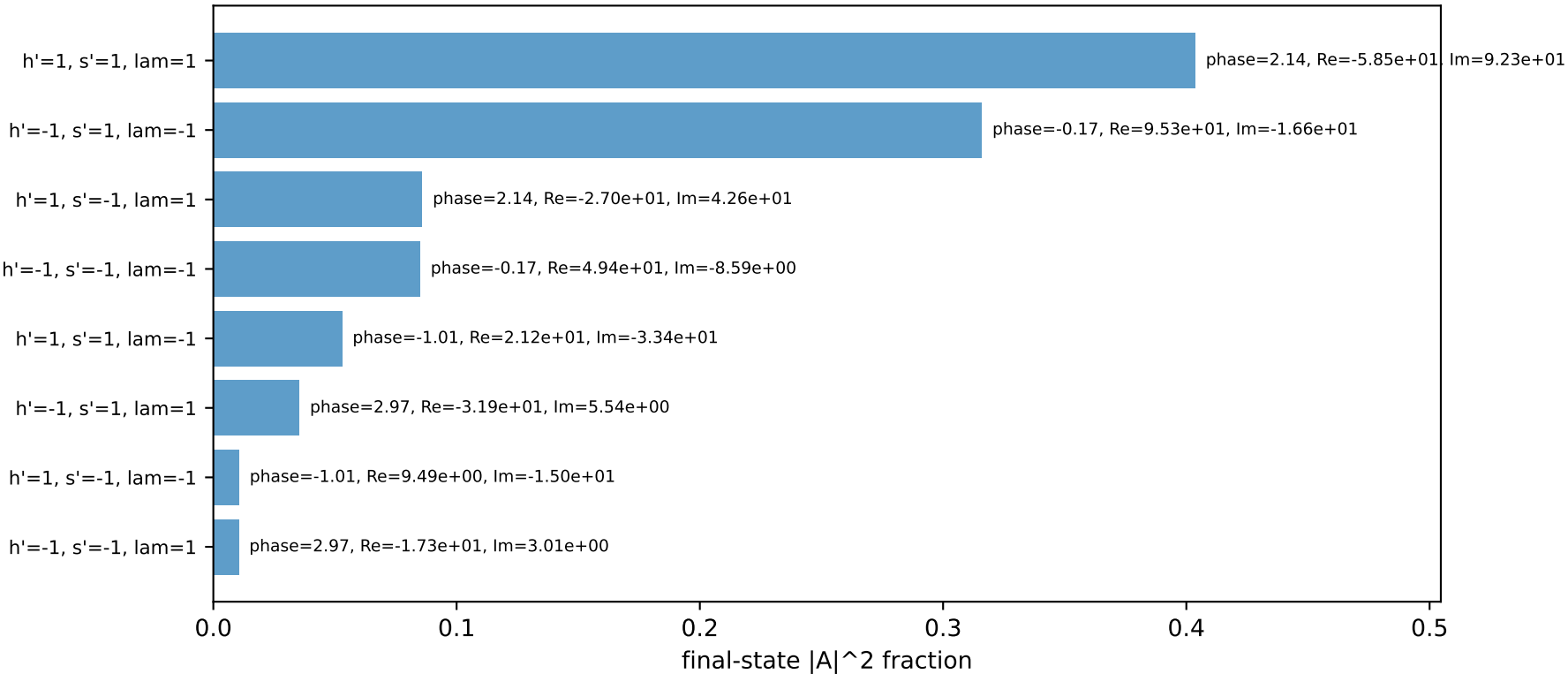
$s=16.7824$, $\sqrt{s}=4.09663$, $p_{In}=1.94093$, $p_{Out}=1.90594$
 $\theta_{in}=1.57079$, $\phi_{e\text{ in}}=5.10509$, $\phi_{p\text{ in}}=1.9635$
 $q_{Out}=1.25$, $\phi_{\gamma}=4.90874$
 $Q^2=0.727804$, $x_B=0.0679444$, $t=-0.56342$, $W^2=10.8638$, $y=0.673589$
 $\theta(e',\gamma)=30.9799$ deg

four-momenta [E, p_x , p_y , p_z] GeV:
k E= 1.9409 $p_x= 0.7428$ $p_y= -1.7932$ $p_z= -0.0000$
p E= 2.1557 $p_x= -0.7428$ $p_y= 1.7932$ $p_z= 0.0000$
kp E= 0.7224 $p_x= -0.2439$ $p_y= -0.6800$ $p_z= 0.0000$
pp E= 2.1243 $p_x= 0.0000$ $p_y= 1.9059$ $p_z= 0.0000$
qout E= 1.2500 $p_x= 0.2439$ $p_y= -1.2260$ $p_z= 0.0000$

Transverse plane

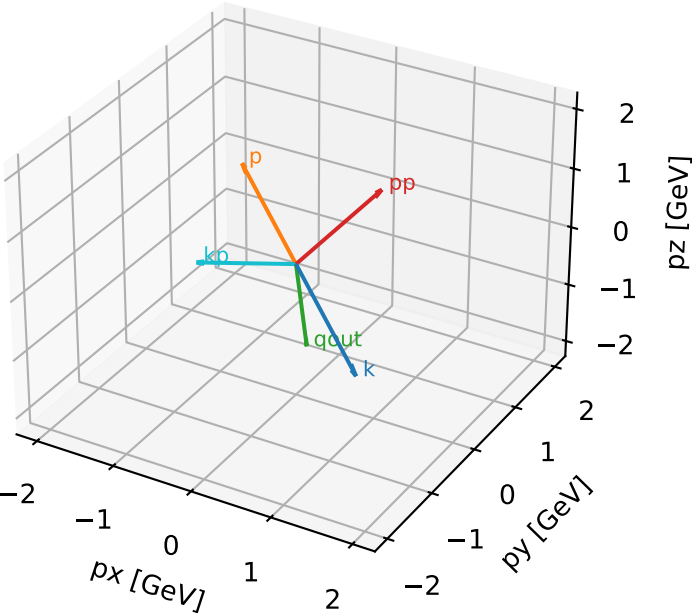


Final-state amplitude decomposition



Ty characteristic high-C13 configuration

3D momenta

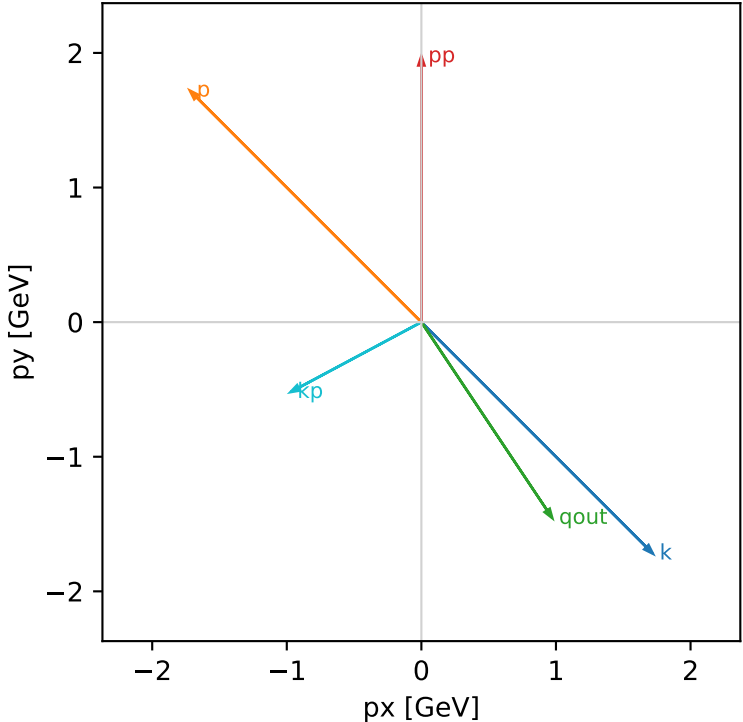


Ty_aligned_cluster_2 C13=0.772624
 region: aligned $|\phi_{e\text{ in}} - \phi_{\gamma}|=0.19635$
 kinematic point: high_s_low theta_in_high_Egamma
 incoming state: $(A[h_{\text{In}}=+1, s_{\text{In}}=1] + i A[h_{\text{In}}=-1, s_{\text{In}}=1]) / \sqrt{2}$

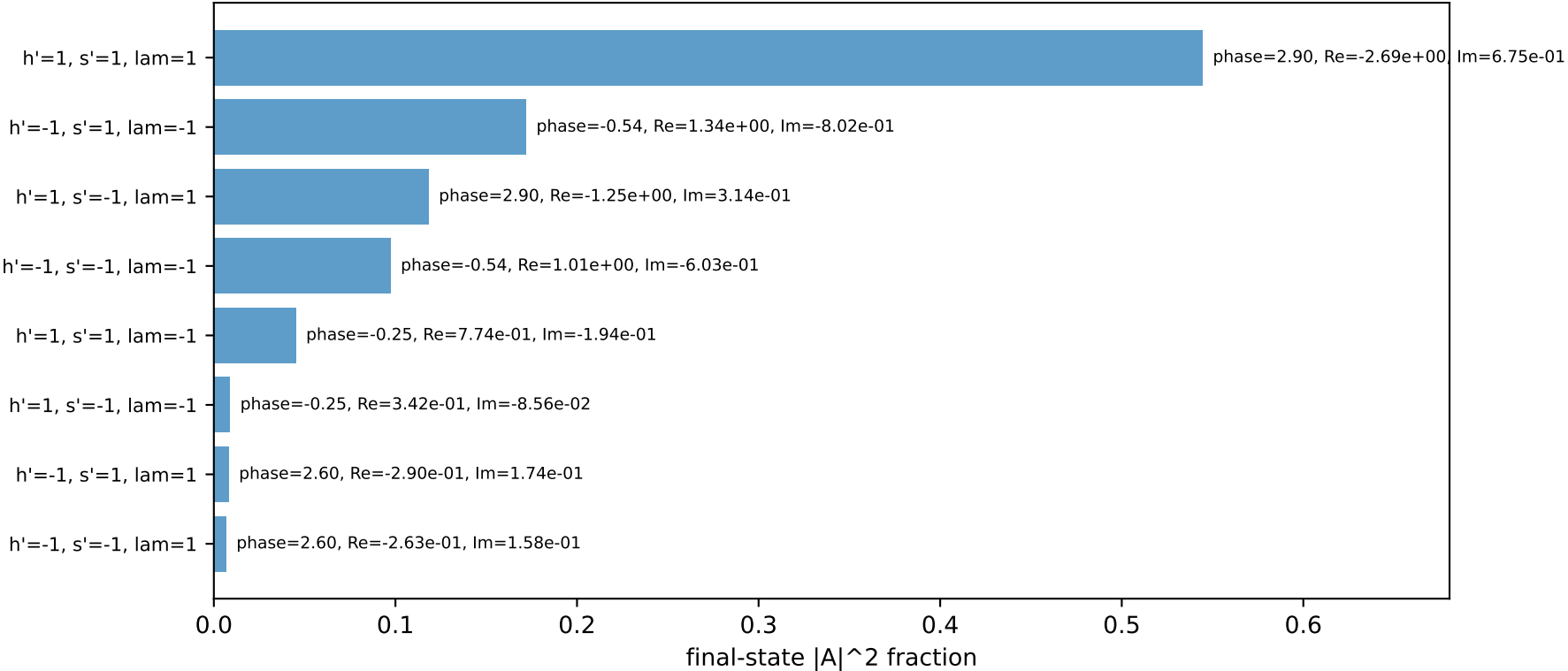
$s=25.3887$, $\sqrt{s}=5.03872$, $p_{\text{In}}=2.43205$, $p_{\text{Out}}=1.97482$
 $\theta_{\text{in}}=1.57079$, $\phi_{e\text{ in}}=5.49779$, $\phi_{p\text{ in}}=2.35619$
 $q_{\text{Out}}=1.75$, $\phi_{\gamma}=5.30144$
 $Q^2=6.91879$, $x_B=0.34053$, $t=-2.84578$, $W^2=14.2788$, $y=0.828996$
 $\theta(e', \gamma)=95.6217$ deg

four-momenta [E, px, py, pz] GeV:
 k E= 2.4321 px= 1.7197 py= -1.7197 pz= -0.0000
 p E= 2.6067 px= -1.7197 py= 1.7197 pz= 0.0000
 kp E= 1.1025 px= -0.9722 py= -0.5197 pz= 0.0000
 pp E= 2.1863 px= 0.0000 py= 1.9748 pz= 0.0000
 qout E= 1.7500 px= 0.9722 py= -1.4551 pz= 0.0000

Transverse plane

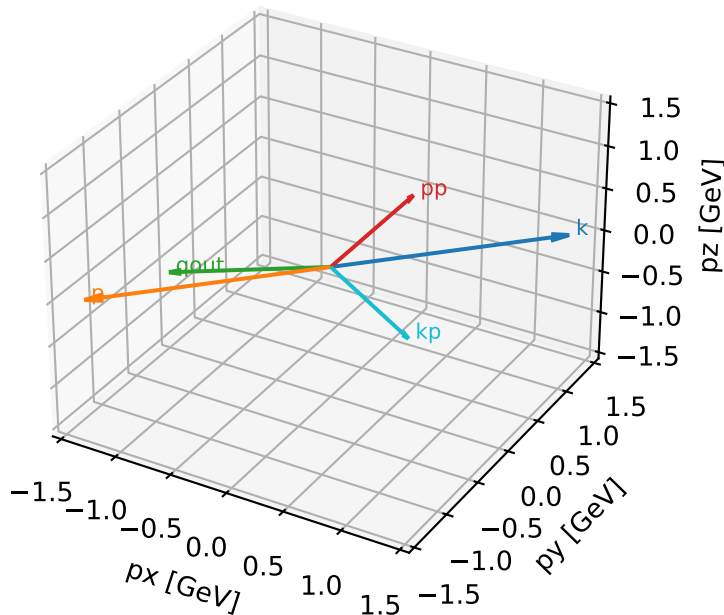


Final-state amplitude decomposition



Ty characteristic high-C13 configuration

3D momenta

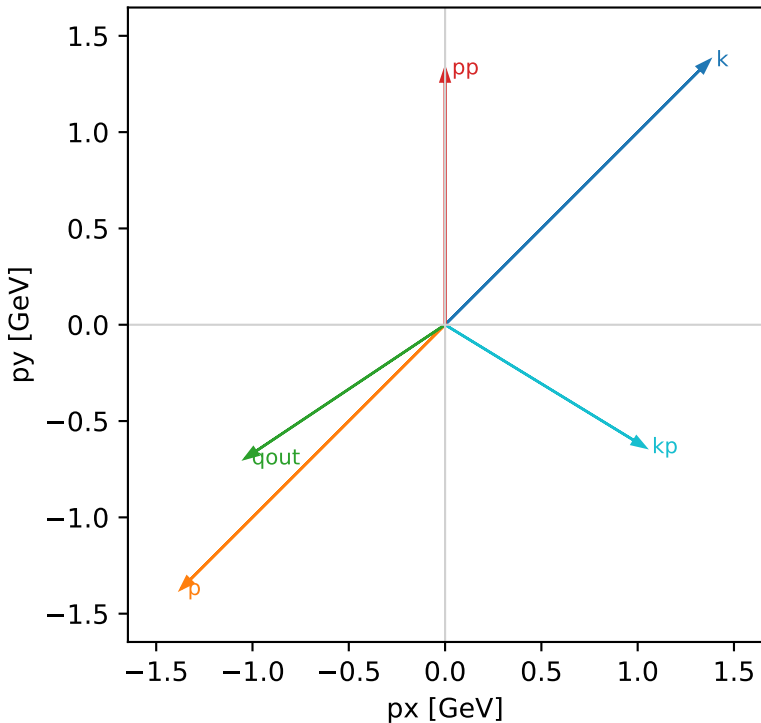


Ty_opposite_cluster_0 C13=0.0590558
region: opposite $|\phi_{e\text{ in}} - \phi_{\gamma}|=2.94524$
kinematic point: low s low θ_{in} mid E_{γ}
incoming state: $(A[h_{\text{In}}=+1, s_{\text{In}}=1] + i A[h_{\text{In}}=-1, s_{\text{In}}=1]) / \sqrt{2}$

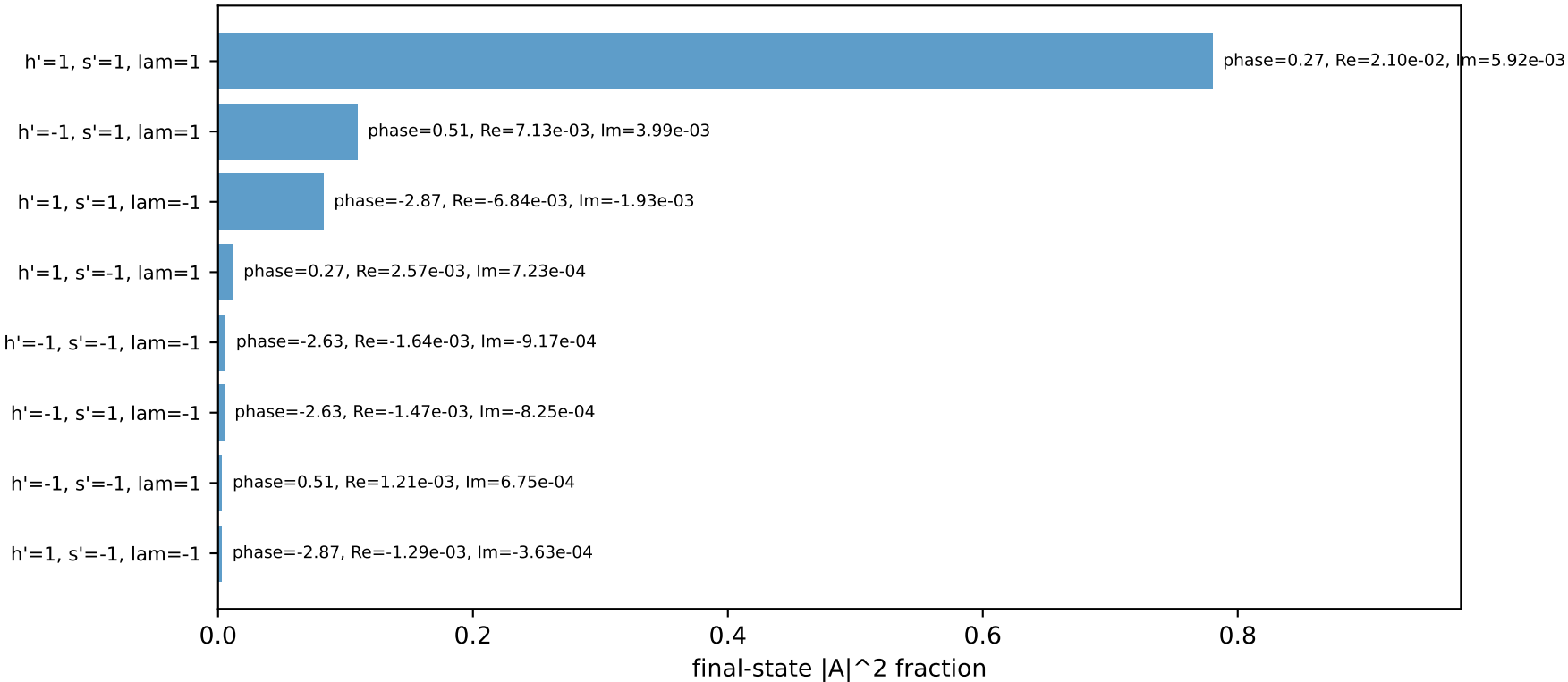
$s=16.7824$, $\sqrt{s}=4.09663$, $p_{\text{In}}=1.94093$, $p_{\text{Out}}=1.33066$
 $\theta_{\text{in}}=1.57079$, $\phi_{e\text{ in}}=0.785398$, $\phi_{p\text{ in}}=3.92699$
 $q_{\text{Out}}=1.25$, $\phi_{\gamma}=3.73064$
 $Q^2=3.62382$, $x_B=0.379772$, $t=-8.91193$, $W^2=6.79812$, $y=0.600037$
 $\theta(e', \gamma)=114.778$ deg

four-momenta [E, p_x , p_y , p_z] GeV:
k E= 1.9409 $p_x= 1.3724$ $p_y= 1.3724$ $p_z= -0.0000$
p E= 2.1557 $p_x= -1.3724$ $p_y= -1.3724$ $p_z= 0.0000$
kp E= 1.2186 $p_x= 1.0393$ $p_y= -0.6362$ $p_z= 0.0000$
pp E= 1.6280 $p_x= 0.0000$ $p_y= 1.3307$ $p_z= 0.0000$
qout E= 1.2500 $p_x= -1.0393$ $p_y= -0.6945$ $p_z= 0.0000$

Transverse plane

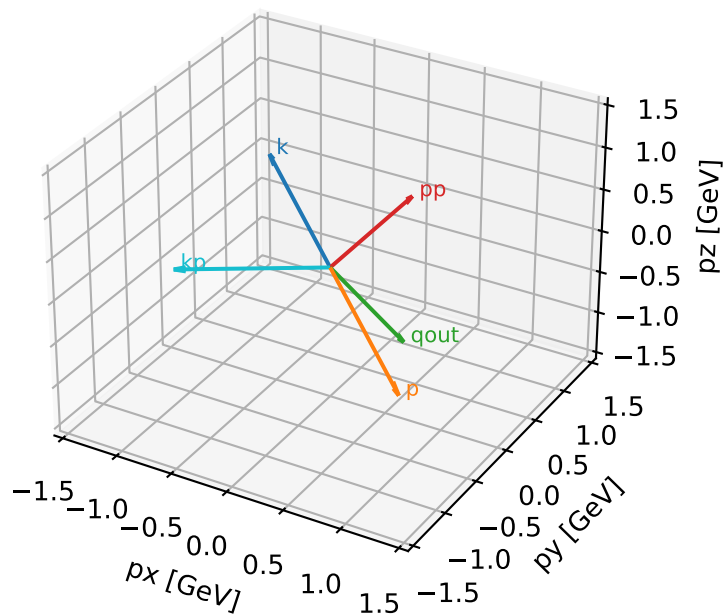


Final-state amplitude decomposition



Ty characteristic high-C13 configuration

3D momenta

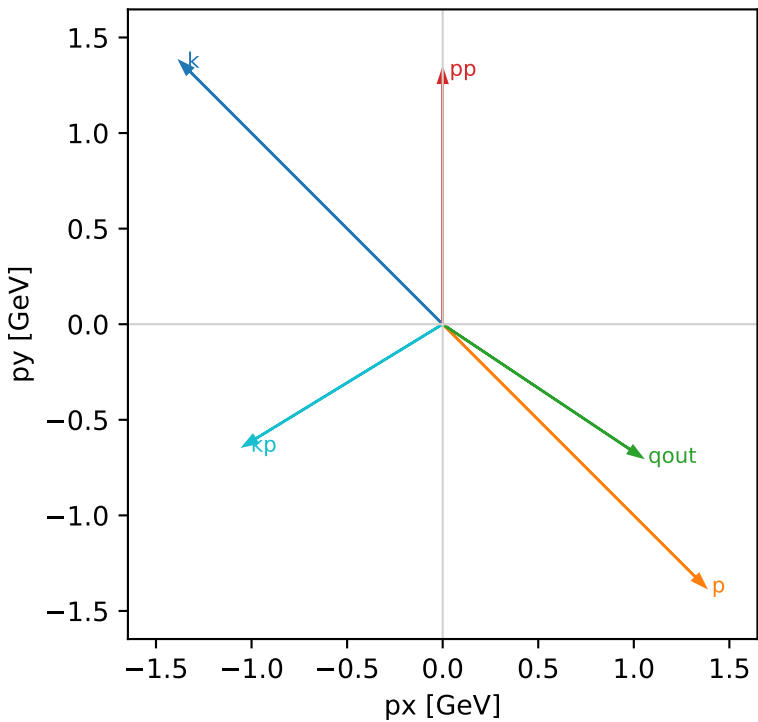


Ty_opposite_cluster_1 C13=0.0590558
 region: opposite $|\phi_{e\text{ in}} - \phi_{\gamma}|=2.94524$
 kinematic point: low s low θ_{in} mid E_{γ}
 incoming state: $(A[h_{\text{In}}=+1, s_{\text{In}}=1] + i A[h_{\text{In}}=-1, s_{\text{In}}=1]) / \sqrt{2}$

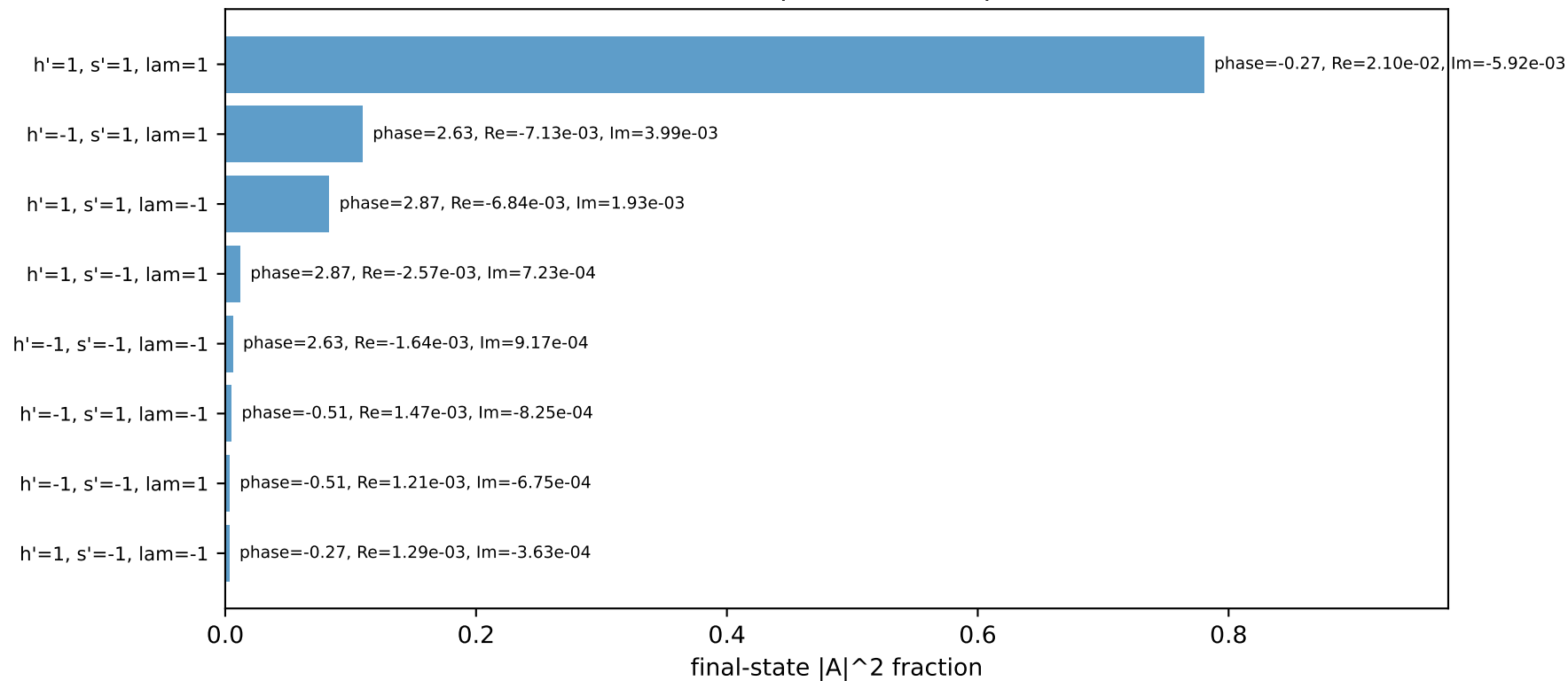
$s=16.7824$, $\sqrt{s}=4.09663$, $p_{\text{In}}=1.94093$, $p_{\text{Out}}=1.33066$
 $\theta_{\text{in}}=1.57079$, $\phi_{e\text{ in}}=2.35619$, $\phi_{p\text{ in}}=5.49779$
 $q_{\text{Out}}=1.25$, $\phi_{\gamma}=5.69414$
 $Q^2=3.62382$, $x_B=0.379772$, $t=-8.91193$, $W^2=6.79812$, $y=0.600037$
 $\theta(e', \gamma)=114.778$ deg

four-momenta [E, p_x , p_y , p_z] GeV:
 k E= 1.9409 p_x = -1.3724 p_y = 1.3724 p_z = -0.0000
 p E= 2.1557 p_x = 1.3724 p_y = -1.3724 p_z = 0.0000
 kp E= 1.2186 p_x = -1.0393 p_y = -0.6362 p_z = 0.0000
 pp E= 1.6280 p_x = 0.0000 p_y = 1.3307 p_z = 0.0000
 qout E= 1.2500 p_x = 1.0393 p_y = -0.6945 p_z = 0.0000

Transverse plane

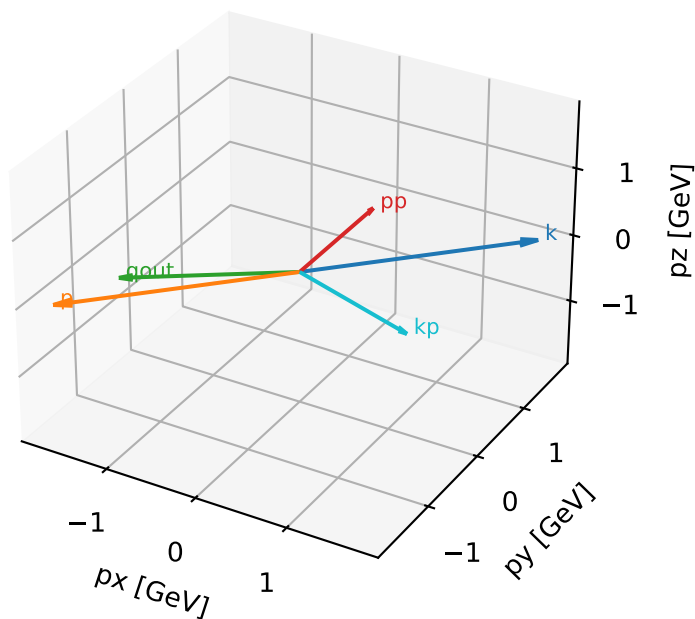


Final-state amplitude decomposition



Ty characteristic high-C13 configuration

3D momenta

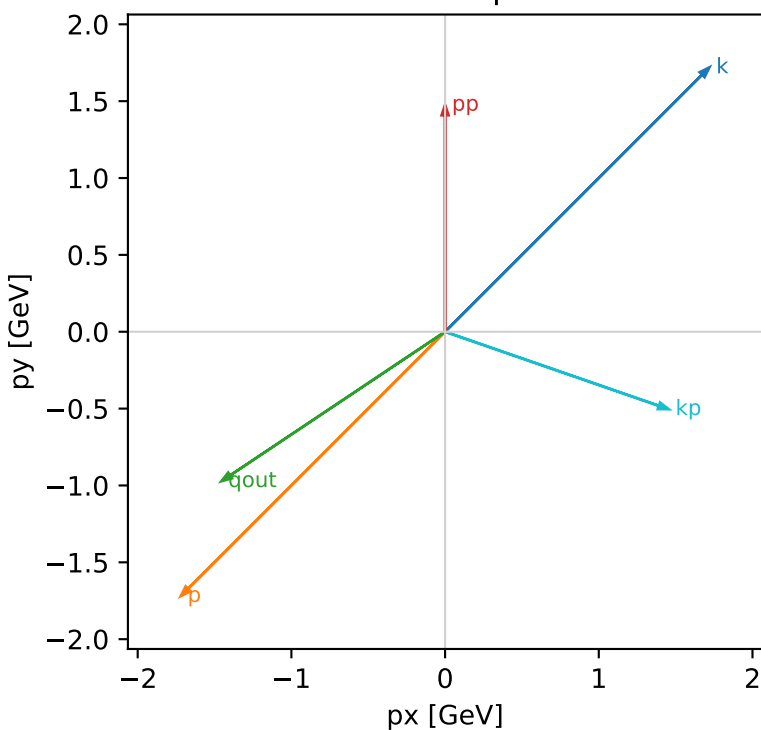


Ty_opposite_cluster_2 C13=0.0554606
 region: opposite $|\phi_{e\ in} - \phi_{\gamma}|=2.94524$
 kinematic point: high_s_low_theta_in_high_Egamma
 incoming state: $(A[h_{In}=+1, s_{In}=1] + i A[h_{In}=-1, s_{In}=1]) / \sqrt{2}$

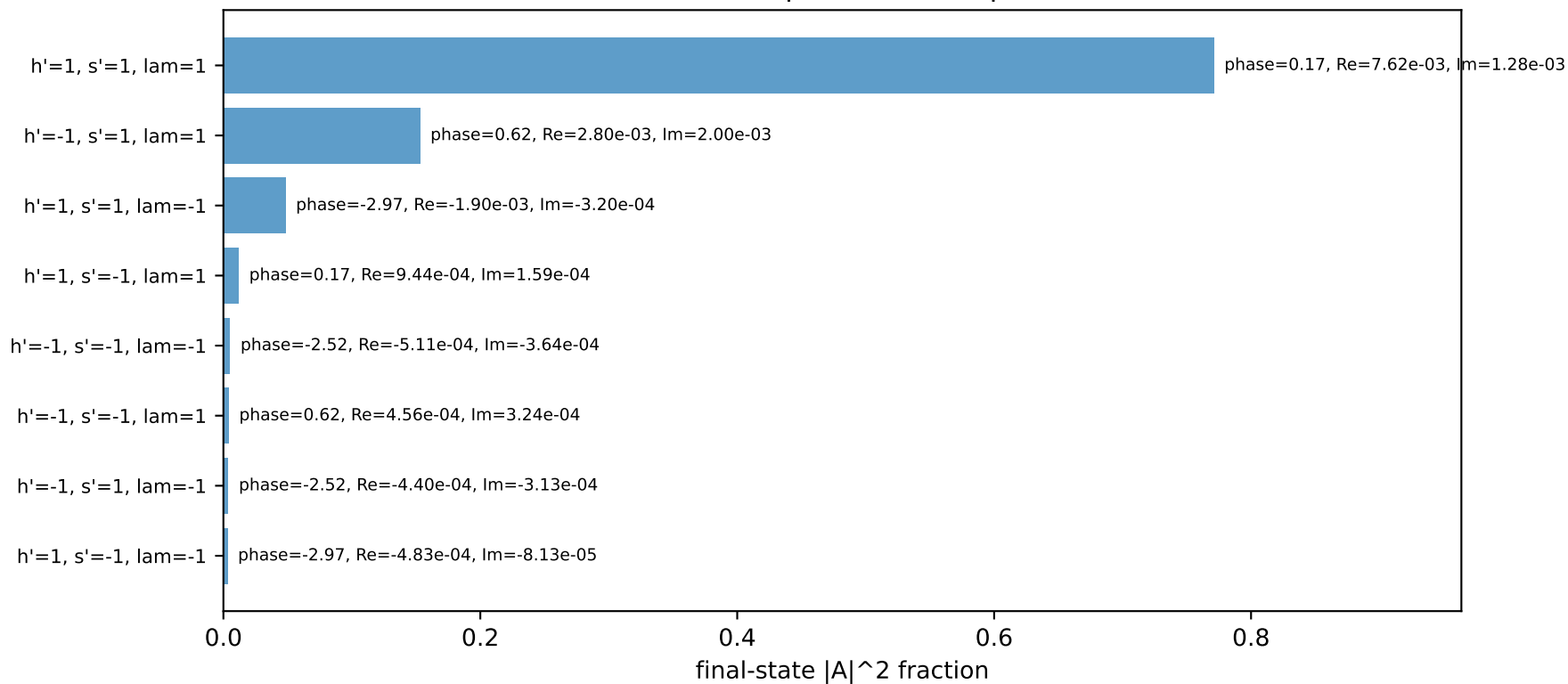
$s=25.3887$, $\sqrt{s}=5.03872$, $p_{In}=2.43205$, $p_{Out}=1.47607$
 $\theta_{in}=1.57079$, $\phi_{e\ in}=0.785398$, $\phi_{p\ in}=3.92699$
 $q_{Out}=1.75$, $\phi_{\gamma}=3.73064$
 $Q^2=4.21811$, $x_B=0.319325$, $t=-12.4347$, $W^2=9.87118$, $y=0.538966$
 $\theta(e', \gamma)=127.152$ deg

four-momenta [E, p_x , p_y , p_z] GeV:
 k E= 2.4321 $p_x= 1.7197$ $p_y= 1.7197$ $p_z= -0.0000$
 p E= 2.6067 $p_x= -1.7197$ $p_y= -1.7197$ $p_z= 0.0000$
 kp E= 1.5398 $p_x= 1.4551$ $p_y= -0.5038$ $p_z= 0.0000$
 pp E= 1.7489 $p_x= 0.0000$ $p_y= 1.4761$ $p_z= 0.0000$
 qout E= 1.7500 $p_x= -1.4551$ $p_y= -0.9722$ $p_z= 0.0000$

Transverse plane

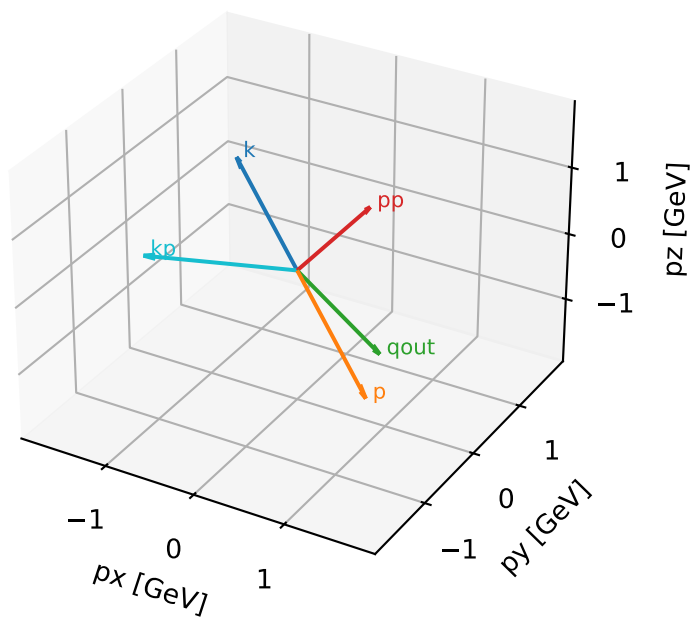


Final-state amplitude decomposition



Ty characteristic high-C13 configuration

3D momenta



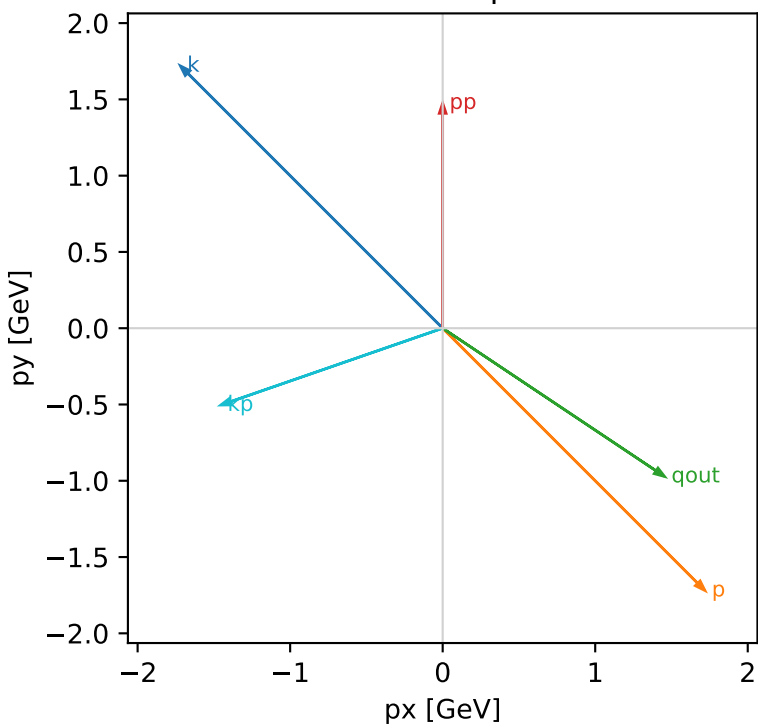
Ty_opposite_cluster_3 C13=0.0554606
 region: opposite $|\phi_{e\text{ in}} - \phi_{\gamma}|=2.94524$
 kinematic point: high_s low_theta_in_high_Egamma
 incoming state: $(A[hIn=+1,sIn=1] + i A[hIn=-1,sIn=1]) / \sqrt{2}$

$s=25.3887$, $\sqrt{s}=5.03872$, $pIn=2.43205$, $pOut=1.47607$
 $\theta_{in}=1.57079$, $\phi_{e\text{ in}}=2.35619$, $\phi_{p\text{ in}}=5.49779$
 $qOut=1.75$, $\phi_{\gamma}=5.69414$
 $Q^2=4.21811$, $x_B=0.319325$, $t=-12.4347$, $W^2=9.87118$, $y=0.538966$
 $\theta(e',\gamma)=127.152$ deg

four-momenta [E, px, py, pz] GeV:

particle	E	px	py	pz
k	2.4321	-1.7197	1.7197	-0.0000
p	2.6067	1.7197	-1.7197	0.0000
kp	1.5398	-1.4551	-0.5038	0.0000
pp	1.7489	0.0000	1.4761	0.0000
qout	1.7500	1.4551	-0.9722	0.0000

Transverse plane



Final-state amplitude decomposition

